1 High-school city evaluation: interim results

Singapore is the conservative recommendation in our current comparison of 25 cities. The model separates a decision made with incomplete information from a claim that one city is certainly best: all 25 cities remain possible score leaders within the available evidence bounds. This section reports the current stage of the city-evaluation study as of 8 November 2026.

1.1 A decision model that retains missing information

The eight main criteria describe safety, particulate air pollution, living cost, housing affordability, upper secondary teaching resources, the supply of higher education institutions, school travel time, and fixed broadband. Listed public transport stops and public swimming pools form two auxiliary supply criteria. Teaching resources measure student–teacher provision; they do not measure learning outcomes. Institution and facility counts measure supply, with partial inventories retained as lower bounds.

Let $u_{ij} \in [\ell_{ij}, h_{ij}] \subseteq [0,1]$ denote city i's utility for criterion j. A completely unidentified value retains $[0,1]$. For a nonnegative measured value x and fixed reference $a > 0$, benefit criteria use $x/(a+x)$ and cost criteria use $a/(a+x)$. The reference values remain fixed across cities. School travel instead uses

$$u_T(t;\tau) = \exp(-t/\tau), \qquad \tau = 30 \text{ minutes}, \tag{1}$$

where t is reported one-way duration. Time bins yield utility intervals; unobserved journeys retain unknown utility. The scale τ expresses the strength of the travel-time penalty and is tested below.

Each main criterion receives weight $(1-\alpha)/8$ and each auxiliary criterion receives $\alpha/2$, with $\alpha = 0.10$ for the reported ranking. We use fixed ideal utilities of zero and one:

$$C_i(u;\alpha) = 100\,\frac{\sqrt{\sum_j w_j(\alpha) u_{ij}^2}}{\sqrt{\sum_j w_j(\alpha) u_{ij}^2} + \sqrt{\sum_j w_j(\alpha)(1-u_{ij})^2}}. \tag{2}$$

The score increases with every utility, so its attainable bounds in the rectangular evidence model are $L_i = C_i(\ell_i;\alpha)$ and $U_i = C_i(h_i;\alpha)$. We rank choices by their worst-case regret:

$$R_i = \max\{0, \max_{k \neq i} U_k - L_i\}, \qquad i^* = \arg\min_i R_i. \tag{3}$$

Thus R_i asks how far choosing city i could leave a student behind a better-scoring alternative. A city remains a possible score leader when $U_i \geq \max_{k \neq i} L_k$. The rectangular model permits input values to vary independently within their bounds; these are conditional decision statements.

1.2 Current ranking and the evidence behind it

Table 1 gives the decision order. Singapore's worst-case regret is 34.07 score points, followed by Tokyo at 37.41. Phoenix ranks 16th. A numerical certificate retains Singapore as the minimum-regret choice for every auxiliary weight $\alpha \in [0.05, 0.20]$ under the current inputs. The overlapping score intervals leave all 25 cities as possible leaders; the decision order therefore expresses a conservative choice under uncertainty.

Table 1: Current minimum-regret ranking; smaller R_i is preferred. Scores use the evidence bounds and $\alpha = 0.10$, $\tau = 30$ minutes.

Rank	City	R_i	Rank	City	R_i
1	Singapore	34.07	14	Delhi	44.94
2	Tokyo	37.41	15	Istanbul	45.09
3	Copenhagen	38.10	16	Phoenix	46.14
4	Melbourne	41.30	17	Athens	46.62
5	Cape Town	42.02	18	London	46.78
6	Shanghai	42.12	19	Jakarta	47.33
7	Sao Paulo	42.67	20	Nairobi	48.24
8	Warsaw	42.82	21	New York	48.42
9	Barcelona	43.30	22	Dhaka	48.49
10	Toronto	43.42	23	Mumbai	48.71
11	Bogota	43.95	24	Mexico City	48.75
12	Dubai	44.05	25	Lagos	51.58
13	Santiago	44.67			

Actual student-travel information currently gives informative, conditional intervals for five cities. New York uses a 2010–11 resident survey of current grades 9–12 with an under-19 age restriction; unknown ages remain optional members. Mexico City uses 2025 private-household residents aged 14–18, with uncertain school-stage membership retained. São Paulo uses 2023 municipal residents aged 14–18 in current upper-secondary education. Cape Town uses 2025 metropolitan residents aged 14–18 in current high-school grades. Melbourne uses 2024–25 Greater Melbourne residents classified as Secondary, aged 15–19. The other 20 cities retain $[0, 1]$ for travel.

These sources differ in year, geography, age coverage and sampling procedure. The São Paulo research mirror has not been verified as byte-identical to the official release. Melbourne's reported interval is conditional on the 170 students with complete person weights; another eligible student's unbounded missing weight leaves the strict full-group interval at $[0, 1]$. Source definitions, utility bounds and ranking inputs accompany this section in the selected statistical snapshot.[1]

[1]Research model V108 and validation V111; reproducible source records and calculations: `https://github.com/TypeThe0ry/HiMCM/tree/1b35f6c/research/2026-a`. The accompanying snapshot is `statistics/task4_progress_2026-11-08/`.

1.3 A revision that accounts for sampling uncertainty

We added sampling uncertainty to Mexico City's 2025 time-bin interval, holding the data and uncertain school-stage membership rules fixed. We retain all 3,433 observed primary sampling units (PSUs) in 49 strata, including 586 units with zero contribution to the target group. For each of nine fixed membership prefixes per endpoint, a Taylor-linearized standard error produces a 97.5% one-sided t bound with 3,384 degrees of freedom; taking the minimum lower bound and maximum upper bound gives an approximate 95% outer envelope for the identified interval. Final nonresponse-adjusted weights are treated as fixed and finite-population corrections are omitted.

Table 2: Mexico City before and after adding sampling uncertainty. The other 249 city–criterion cells are unchanged.

Quantity	Before	After
Travel utility interval	$[0.213824, 0.584100]$	$[0.207414, 0.592051]$
Worst-case regret	48.7028	48.7469
Decision rank	23	24

Mexico City moves down one place and Mumbai up one place; other ranks remain unchanged. This corrects the treatment of uncertainty and leaves 25 possible leaders. School-stage interpretation and coverage remain conditions of the estimate; the approximation has no exact finite-sample coverage guarantee.

1.4 Recommendation stability and rank sensitivity

We re-evaluated the five informative travel sources at five prespecified scales, retaining their original membership, missing-data and sampling rules. Each scale reproduces utilities from the available duration or time-bin information. Kendall's τ_b compares each decision order with the default-scale order.

Table 3: Direct travel-scale sensitivity relative to $\tau = 30$ minutes.

τ (minutes)	15	20	30	45	60
Cities changing rank	11	9	0	9	10
Kendall's τ_b	0.9267	0.9533	1.0000	0.9600	0.9400

Singapore remains the conservative recommendation across all five scales and the continuous α range at each scale. Cape Town ranks fourth to ninth and Mexico City nineteenth to twenty-fourth. All 25 cities remain possible leaders. Thus the recommendation is stable across the five tested scales while intermediate ranks vary. Values between the tested τ settings were not continuously certified.